

Degree Requirements

The Bachelor of Science in Game Design is a four-year program that consists of 121 credits: 40 credits of General Education, 66 credits in Game Design courses (required and elective courses related to a focus area), and 15 credits in General Electives.

Information on admission criteria, including allowable credits for transfer students, is announced yearly on the College of Design website.

General Education

University Requirement

International Perspectives	3
--	---

U.S. Cultures and Communities	3
---	---

Total	<u>6</u>
-------	-----------------

Communications

ENGL 1500 Critical Thinking and Communication*	3
--	---

ENGL 2500 Written, Oral, Visual, and Electronic Composition*	3
--	---

Select 1 course from: COMST 1010, 1020, SP CM 1100, 2120	3
--	---

LIB 1600 Introduction to College Level Research	1
---	---

Total	<u>10</u>
-------	------------------

* With a C or better

<i>Humanities</i>	<u>6</u>
-------------------	-----------------

6 credits from General Education Approved Course List

<i>Social Sciences</i>	<u>6</u>
------------------------	-----------------

6 credits from General Education Approved Course List

<i>Mathematical, Physical & Biological Science</i>	<u>6</u>
--	-----------------

6 credits from General Education Approved Course List

<i>Additional General Education Courses</i>	<u>6</u>
---	-----------------

3000-4000 Level from General Education Approved Course List

Total General Education & Electives	<u>40</u>
--	------------------

Game Design Focus Areas

Game Design majors must choose a Focus Area: Game Art, Game Computing, or Game Worlds. All focus area study plans include a common core of required Game Design courses as well as required courses and electives that are specific to the focus area.

Game Art will prepare students to solve visual problems in gaming and will learn how to utilize this knowledge in the game development process. Students will gain skills in drawing, visual design and storytelling, storyboarding, character animation, 3D modeling, and related fields.

Game Computing will provide comprehensive knowledge in programming, coding, and creating online digital games. Students will focus on object-oriented programming, data structures, and discrete computational structures which they will be able to apply in game design and development.

Game Worlds will provide students with the skills and knowledge to create meaningful and engaging worlds for games of every scale, understand and design dynamic stories, and comprehend the cultural relevance of the games they play, study, and develop. Within Game Worlds, courses are subdivided into Game Worlds, Narrative Design, and Society. This focus area will support students' acquisition of the skills to apply cultural lessons to the writing and narrative design of games.

Game Art Focus Area

Game Art Required

COM S 1270 Introduction to Programming	3
DSN S 1310 Drawing I	4
GAME 2020 Game Design Fundamentals	3
GAME 2030 Introduction to Game Engines	3
ARTIS 2120 Studio Fundamentals: Digital Media	3
ARTIS 2300 Drawing II	3
GAME 2100 Game Career Development	2
GAME 2110 Gameplay and Game Analysis	3
Choose One:	
GAME 2720 Game Design and History	3
GAME 2730 Game Design and Cultures	3
GAME 3010 Game Design Workshop	3
GAME 4010 Game Design Capstone	6

Total Game Art Required	<u>36</u>
--------------------------------	------------------

Game Art Electives

Choose <i>18 credits</i> from the following courses	18
---	-----------

ARTIS 3080 Computer Modeling and Virtual Photography

ARTIS 4060 Introduction to 3D Organic Modeling in Zbrush

ARTIS 4070 Principles of Character Animation

ARTIS 3230 Scientific Illustration Principles and Techniques

ARTIS 3260 Illustration and Illustration Software

ARTIS 3270 Illustration as Communication

ARTIS 4080 Principles of 3D Animation

ARTGR 4840-2 3D Modeling and Augmented Reality

ARTIS 4310 Character and Scene Design

ARTIS 4320 Sequential Narrative Drawing

ARTGR 4630 3D Motion Graphics

ARTIS 4700 Data Coding and Form

ARTIS 4730 Video Art

Total Game Art Electives	<u>18</u>
---------------------------------	------------------

Game Design Exploratory Electives

Choose 6 credits from other Focus Areas from Master List:	6
---	----------

Choose *3 credits* from the Master List for Game Computing

Choose *3 credits* from the Master list for Game Worlds

Choose 6 cr. from Game Art, Game Computing, and/or Game Worlds List	6
---	----------

Total Game Design Exploratory Electives	<u>12</u>
--	------------------

Game Art Focus Area Total	<u>66</u>
----------------------------------	------------------

Game Computing Focus Area

Game Computing Required

COM S 1270 Introduction to Programming	3
DSN S 1310 Drawing I	4
GAME 2020 Game Design Fundamentals	3
GAME 2030 Introduction to Game Engines	3
GAME 2040 Programming for Game Engines	3
ARTIS 2120 Studio Fundamentals: Digital Media	3
GAME 2100 Game Career Development	2
GAME 2110 Gameplay and Game Analysis	3
Choose One:	
GAME 2720 Game Design and History	3
GAME 2730 Game Design and Cultures	3
GAME 3010 Game Design Workshop	3
GAME 4010 Game Design Capstone	6
COM S 2270 Object Oriented Programming	3
COM S 2280 Introduction to Data Structures	3
COM S 2300 Discrete Computational Structures	3
COM S 3110 Introduction to the Design and Analysis of Algorithms	3
COM S 3270 Advanced Programming Techniques	3
COM S 3360 Introduction to Computer Graphics	3
COM S 4370 Computer Game and Media Programming	3
Total Game Computing Required	<u>57</u>

Game Computing Electives

Choose 3 credits from the following courses	3
COM S 3360 Computer Graphics	

CYB E 231 Cyber Security Concepts and Tools

CRP 3010 Urban Analytical Methods

CRP 4560 GIS Programming and Automation

EDUC 3020 Principles and Practices of Learning with Technology

S E 3090 Software Development Practices

S E 3170 Introduction to Software Testing

S E 3190 Construction of User Interfaces

S E 3390 Software Architecture and Design

S E 4210 Software Analysis and Verification for Safety and Security

Total Game Computing Electives **3**

Game Design Exploratory Electives

Choose 6 credits from the Master List for Game Art or Game Worlds **6**

Total Game Design Exploratory Electives **6**

Game Computing Total **66**

Game Worlds Focus Area

Game Worlds Required

COM S 1270 Introduction to Programming	3
DSN S 1310 Drawing I	4
GAME 2020 Game Design Fundamentals	3
GAME 2030 Introduction to Game Engines	3
ARTIS 2120 Studio Fundamentals: Digital Media	3
GAME 2100 Game Career Development	2
GAME 2110 Gameplay and Game Analysis	3
Choose One:	
GAME 2720 Game Design and History	3
GAME 2730 Game Design and Cultures	3
GAME 3010 Game Design Workshop	3
GAME 4010 Game Design Capstone	6
Choose 3 credits from the following courses:	3
ENGL 3040 Creative Writing: Fiction	
ENGL 3150 Creative Writing: Screenplays	
ENGL/THTRE 3160 Creative Writing: Playwriting	
Total Game Worlds Required	<u>36</u>

Game Worlds Electives

Choose 3 credits from the following courses (World Building):	3
ANTHR 2300 Globalization and the Human Condition	
HIST 2840 Wonders of the World, Global Innovation	
POL S 1250 Democracy and Dictatorship	
RELIG 2050/WLC 2050 World Religions	
CRP 2010 The North American Metropolis	

Choose 3 credits from the following courses (Narrative Design): 3

CL ST 2730 Greek and Roman Mythology

ENGL 2750 Analysis of Popular Cultural Texts

PHIL 3630 Metaphysics in Science Fiction and Popular Culture

Choose 3 credits from the following courses (Society): 3

ART H 2800/2810 History of Art I/II

HIST 2710 History of Sports in U.S.

RELIG 2150 Religion and Popular Culture

PHIL 2300 Moral Theory and Practice

CRP 2910 World Cities and Globalization

Choose 3 credits from the following courses (Open Topics): 3

AM IN 3220 Peoples and Cultures of Native North America

ART H 3950 Art and Theory Since 1945

CRP 3200 Urban Geography

ENGL 3150 Creative Writing: Screenplays

ENGL 3300 Science Fiction

HIST 3040 Cultural Heritage of the Ancient World

HIST 3640 The Mythic Wild West

PHIL 3640 Metaphysics: God, Minds, and Matter

POL S 308X Video Games and Politics

POL S 3340/SOC 3340 Politics and Society

Choose 6 credits from the following courses (Open Topics): 6

AF AM 3300/SOC 3300 Ethnic and Race Relations

ENGL 3020 Business Communication

ENGL 3040 Creative Writing Fiction

ENGL 3130 Rhetorical Web Design

ENGL/THE 3160 Creative Writing: Playwriting

ART H 4940/WGS 4940 Women and Gender in Art
HIST 4880 American Stuff, Colonial Times to the Present
PHIL 3430 Philosophy of Technology
PSYCH 3860 Media Psychology
WGS 4300 Gender and Consumer Culture

Total Game Worlds Electives **18**

Game Design Exploratory Electives

Choose 6 credits from other Focus Areas from Master List: 6

Choose 3 *credits* from the Master List for Game Computing

Choose 3 *credits* from the Master list for Game Art

Choose 6 cr. from Game Art, Game Computing, and/or Game Worlds List 6

Total Game Design Major Electives **12**

Game Worlds Total **66**

General University Electives **15**

Game Design Course Master List

This master list contains courses required or recommended by the Game Design Major (excluding Common Core courses). Refer to this list when choosing required courses from other focus areas and other optional courses.

Game Design (General)

GAME 2040 – Programming for Game Engines

GAME 2101- Game Design Field Study

GAME 2102 - Game Design Internship

Game Art Focus Area

ARTIS 2300 - Drawing II

ARTIS 3080 - Computer Modeling, Rendering and Virtual Photography

ARTIS 3230 - Scientific Illustration Principles and Techniques

ARTIS 3260 - Illustration and Illustration Software

ARTIS 3270 - Illustration as Communication

ARTIS 4060X - Introduction to 3D Organic Modeling in ZBrush

ARTIS 4070 - Principles of Character Animation

ARTIS 4080 - Principles of 3D Animation

ARTIS 4310 - Character and Scene Design

ARTIS 4320 - Sequential Narrative Drawing

ARTIS 4700X - Data Coding and Form

ARTIS 4730 - Video Art

ARTIS 4750 - Interactive Art

ARTGR 4840 2 - 3D Modeling and Augmented Reality

ARTGR 4630 - 3D Motion Graphics

Game Computing Focus Area

GAME 2040 Programming for Game Engines

COM S 2270 - Object Oriented Programming

COM S 2280 - Introduction to Data Structures

COM S 2300 - Discrete Computational Structures
LA 2110 - Digital Design Methods for Landscape Architecture
C R P 2510 - Fundamentals of Geographic Information Systems
COM S 3110 - Introduction to the Design and Analysis of Algorithms
COM S 3270 - Advanced Programming Techniques
COM S 3360 - Computer Graphics
CRP 3510 - Intermediate GIS
COM S 4370 - Computer Game and Media Programming
C R P 3010 - Urban Analytical Methods
C R P 4560 - GIS Programming and Automation
CRP 4490 - Geodesign: Planning for Sustainable Futures
CYB E 2310 - Cyber Security Concepts and Tools (qualifies as UD)
EDUC 3020 - Principles and Practices of Learning with Technology
S E 3090 - Software Development Practices
S E 3170 - Introduction to Software Testing
S E 3190 - Construction of User Interfaces
S E 3390 - Software Architecture and Design
S E 4210 - Software Analysis and Verification for Safety and Security

Game Worlds Focus Area

ANTHR 2300 - Globalization and the Human Condition
ART H 2800/2810 - History of Art I/II
CL ST 2730 - Greek and Roman Mythology
C R P 2010 - The North American Metropolis
C R P 2910 - World Cities and Globalization
ENGL 2750 - Analysis of Popular Cultural Texts
ENGL 3020 - Business Communication
ENGL 3130 - Rhetorical Web Design

ENGL 3140 - Technical Communication

HIST 2710 - History of Sports in U.S.

PHIL 2300 - Moral Theory and Practice

POL S 1250 - Democracy and Dictatorship: Introduction to Comparative Politics

RELIG 205/WLC 2050 - World Religions

RELIG 2150 - Religion and Popular Culture

HIST 2840 - Wonders of the World, Global Innovation

AF AM 330/SOC 3300 - Ethnic and Race Relations

AM IN 3220 - Peoples and Cultures of Native North America

ART H 3950 - Art and Theory Since 1945

ART H 4940/WGS 4940 - Women and Gender in Art

CRP 3200 - Urban Geography

CRP 4550 - Smart and Sustainable Cities

ENGL 3040 - Creative Writing: Fiction

ENGL 3150 - Creative Writing: Screenplays

ENGL 3300 - Science Fiction

ENGL 4110 - Technology, Rhetoric, and Professional Communication

HIST 3040 - Cultural Heritage of the Ancient World

HIST 3640 - The Mythic Wild West

HIST 4880 - American Stuff, Colonial Times to the Present

PHIL 3430 - Philosophy of Technology

PHIL 3630 - Metaphysics in Science Fiction and Popular Culture

PHIL 3640 - Metaphysics: God, Minds, and Matter

POL S 3080X - Video Games and Politics

POL S 3340/SOC 3340 - Politics and Society

PSYCH 3860 - Media Psychology

WGS 4300 - Gender and Consumer Culture

